AN UPPER BOUND FOR THE BOHR RADIUS OF THE UNIT BALL OF ℓ_p^n, $1 < p < 2$

ØISTEIN SØVIK

Abstract. Let $K(B_p^n)$ denote the Bohr radius of the open unit ball of ℓ_p^n. It is known that $K(B_p^n)$ is of order $((\log n)/n)^{1-1/p}$ for $1 \le p \le 2$ and that $\limsup_{n\to\infty} K(B_p^n)/((\log n)/n)^{1-1/p} \le 1$. Defant, García, Maestre and Sevilla-Peris asked whether this limit superior is in fact a limit equal to 1 for $1 < p < 2$, as it is for $2 \le p \le \infty$. We show that the answer is negative for every $1 < p < 2$: the limit superior is at most $\exp(-(1 - s + s\log s)/2) < 1$, where $s = 2 - 2/p$. The proof uses random tetrahedral polynomials and keeps the pointwise variance in the Kahane–Salem–Zygmund argument.

1. Introduction

For a Reinhardt domain $R \subset \mathbb{C}^n$, let $H_\infty(R)$ denote the space of bounded holomorphic functions $f(z) = \sum_\alpha c_\alpha(f) z^\alpha$ on R, with $\|f\|_\infty = \sup_{z\in R}|f(z)|$. The Bohr radius of R is

$$K(R) = \sup\Big\{0 \le r \le 1 : \sup_{z\in rR} \sum_\alpha |c_\alpha(f) z^\alpha| \le \|f\|_\infty \text{ for all } f \in H_\infty(R)\Big\}.$$

Bohr [5] showed that $K(\mathbb{D}) = 1/3$. We are concerned with the open unit balls B_p^n of ℓ_p^n, $1 \le p \le \infty$, and we put

$$h_{p,n} = \left(\frac{\log n}{n}\right)^{1-1/\min\{p,2\}}.$$

The order of $K(B_p^n)$ is known. For the upper bounds, the case $p = \infty$ was discussed by Dineen and Timoney [11] and clarified by Boas and Khavinson [4], the case $p = 1$ is due to Aizenberg [1], and the general case to Boas [3, Theorem 3]. Boas suggested that the logarithmic factor in his bounds is an artifact of the proof, and Defant and Frerick [6] showed that it is not. The lower bound $K(B_\infty^n) \gtrsim h_{\infty,n}$ is due to Defant, Frerick, Ortega-Cerdà, Ounaïes and Seip [8], and by the comparison $K(B_\infty^n) \le K(R)$ of Boas and Khavinson [4] it extends to $2 \le p \le \infty$. Defant and Frerick [7] proved the lower bound for $1 \le p < 2$. In summary [9, Theorem 19.1], $K(B_p^n) \asymp h_{p,n}$ for

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every $1 \le p \le \infty$. Bayart, Pellegrino and Seoane-Sepúlveda [2] then found the exact asymptotics $K(B_\infty^n)/h_{\infty,n} \to 1$.

In their monograph, Defant, García, Maestre and Sevilla-Peris prove that [9, (19.1)]

$$\limsup_{n\to\infty} \frac{K(B_p^n)}{h_{p,n}} \le 1, \qquad 1 \le p \le \infty, \tag{1}$$

and combine it with the result of Bayart, Pellegrino and Seoane-Sepúlveda and the comparison with the polydisc to obtain [9, (19.2)]

$$\lim_{n\to\infty} \frac{K(B_p^n)}{h_{p,n}} = 1, \qquad 2 \le p \le \infty. \tag{2}$$

They write [9, p. 487]: "Whether or not the equality from (19.2) holds for $1 < p < 2$ is unknown". The case $p = 1$ is excluded for a simple reason: then $h_{1,n} = 1$, while restricting to functions of z_1 alone gives $K(B_1^n) \le K(\mathbb{D}) = 1/3$.

The main purpose of the present paper is to answer this question in the negative.

Theorem 1. *Suppose that $1 < p < 2$, and put $s = 2 - 2/p$. Then*

$$\limsup_{n\to\infty} \frac{K(B_p^n)}{((\log n)/n)^{1-1/p}} \le \kappa_p, \qquad \textit{where} \qquad \kappa_p = \exp\left(-\frac{1 - s + s\log s}{2}\right).$$

Since $0 < s < 1$ and the function $s \mapsto 1 - s + s\log s$ has derivative $\log s < 0$ on $(0, 1)$ and vanishes at $s = 1$, we have $\kappa_p < 1$. In particular, (2) fails for every $1 < p < 2$. As $p \to 2$ we have $\kappa_p \to 1$, in accordance with (2) for $p = 2$. Combined with the lower bound of Defant and Frerick [7], Theorem 1 shows that the ratio $K(B_p^n)/((\log n)/n)^{1-1/p}$ has a positive limit inferior and a limit superior at most $\kappa_p < 1$. To the best of our knowledge, Theorem 1 is new.

Let us briefly describe our approach, in the language of [9, Chapter 19]. Since $K(B_p^n) \le K_m(B_p^n)$ for every m [9, Proposition 19.4], it suffices to find, for suitable m, one m-homogeneous polynomial P and one point $z \in B_p^n$ at which the majorant $\sum_\alpha |c_\alpha(P) z^\alpha|$ is large compared with $\|P\|_\infty$. By [9, Lemma 19.5], this amounts to a lower bound for the unconditional basis constant of the monomials in $\mathcal{P}_m(\ell_p^n)$. As in the proof of (1) in [9, Section 19.4], P is a random polynomial and z is the flat point $n^{-1/p}(1, \dots, 1)$. We use the tetrahedral polynomials

$$P_\varepsilon(z) = \sum_{|S|=m} \varepsilon_S z^S, \qquad z^S = \prod_{j\in S} z_j,$$

where S ranges over the subsets of $\{1, \dots, n\}$ with m elements and $\varepsilon_S = \pm 1$. In the notation of [9], $z^S = z_{\mathbf{j}}$ for the index $\mathbf{j} \in \mathcal{J}(m, n)$ that lists the elements of S in increasing order.

Two properties of P_ε matter. First, its majorant is largest at the flat point: by Maclaurin's inequality and Hölder's inequality,

$$\sum_{|S|=m} |z^S| \le \binom{n}{m} \Big(\frac{1}{n}\sum_{j=1}^{n} |z_j|\Big)^m \le \binom{n}{m} n^{-m/p}, \qquad z \in B_p^n,$$

with equality at $z = n^{-1/p}(1,\dots,1)$. Second, the Kahane–Salem–Zygmund argument of [9, Section 17.1] with exponent $r = 2$ gives signs for which $\|P_\varepsilon\|_\infty$ is at most a multiple of $\sqrt{n \log m}$ times the largest pointwise standard deviation

$$V = \sup_{z \in B_p^n} \Big(\sum_{|S|=m} |z^S|^2 \Big)^{1/2}.$$

The estimate $\|z\|_2 \le 1$ gives only $V^2 \le 1/m!$, and with this bound the argument gives only an upper bound for $K(B_p^n)$ of order $m^{1/2} n^{-(1-1/p)}$, which exceeds $h_{p,n}$ by a power of $\log n$. For $p < 2$, however, the majorant and the variance are large at different points: the majorant when the ℓ_p mass is spread over all n coordinates, the variance when it sits on a number of coordinates comparable to m. Lemma 1 below quantifies this: $V^2 \le m^{-m(2/p-1)}/m!$. For $p = 2$ both are largest at the flat point, and there is nothing to gain.

The gain shows up in the constant. Put $\sigma = 1 - 1/p$. Up to factors whose m-th roots tend to 1 as $n \to \infty$ with $m \asymp \log n$, both the argument of [9, Section 19.4], which uses the coefficients $m!/\alpha!$ and [9, Corollary 17.6], and our argument give

$$K(B_p^n) \le \Big(\frac{m}{n}\Big)^{\sigma} \big(n e^{-m}\big)^{\gamma/m}, \tag{3}$$

with $\gamma = \sigma$ in [9] and $\gamma = 1/2$ here. The factor n comes from the size of the net in the Kahane–Salem–Zygmund argument, and e^{-m} from Stirling's formula for $m!$. Writing $m = t \log n$, the right-hand side of (3) equals $t^\sigma e^{\gamma(1/t-1)} ((\log n)/n)^\sigma$. For $\gamma = \sigma$ this is minimal at $t = 1$, where $n e^{-m} = 1$, and the constant is exactly 1; this is where the constant in (1) comes from. Since $\sigma < 1/2$ for $p < 2$, the larger exponent $\gamma = 1/2$ makes it worthwhile to take m larger than $\log n$, so that $n e^{-m} < 1$: the minimum is at $t = 1/(2\sigma)$, and its value is $\kappa_p < 1$.

Organization. The present paper is organized into three additional sections. In Section 2 we estimate the pointwise variance and state the Kahane–Salem–Zygmund bound in the form we need. In Section 3 we prove Theorem 1. Section 4 contains concluding remarks.

2. The pointwise variance and a random polynomial

Throughout, $1 \le m \le n$, and sums over S run over the subsets of $\{1,\dots,n\}$ with m elements. We first record the variance bound, which is the only point where the assumption $p \le 2$ enters.

Lemma 1. *Suppose that $1 \le p \le 2$ and $z \in B_p^n$. Then*

$$\sum_{|S|=m} |z^S|^2 \le \frac{m^{-m(2/p-1)}}{m!}.$$

Proof. Put $x_j = |z_j|^p$, so that $\sum_j x_j \le 1$. For $|S| = m$, the inequality of arithmetic and geometric means gives

$$\prod_{j\in S} x_j \le \Big(\frac{1}{m}\sum_{j\in S} x_j\Big)^m \le m^{-m}.$$

$2/p - 1 \ge 0$, we get

$$|z^S|^2 = \Big(\prod_{j\in S} x_j\Big)^{2/p-1} \prod_{j\in S} x_j \le m^{-m(2/p-1)} \prod_{j\in S} x_j.$$

Finally, when $(x_1 + \cdots + x_n)^m$ is expanded, each product $\prod_{j\in S} x_j$ with $|S| = m$ occurs $m!$ times, and all other terms are nonnegative. Hence $\sum_S \prod_{j\in S} x_j \le (\sum_j x_j)^m/m! \le 1/m!$. □

The bound in Lemma 1 is much larger than the variance at the flat point, which is $\binom{n}{m} n^{-2m/p} \le n^{-m(2/p-1)}/m!$. At points with m non-zero coordinates, all equal to $m^{-1/p}$, the variance is $m^{-2m/p}$, which is smaller than the bound only by the factor $m^m/m! \le e^m$.

The next result is the proof of [9, Theorem 17.3] with exponent $r = 2$, except that the pointwise variance is kept instead of being estimated through the ℓ_2 norm of z. We include the short proof, with explicit constants, to keep the paper self-contained; only the order of the constant matters in the proof of Theorem 1.

Lemma 2. *Suppose that $\|\cdot\|$ is a norm on $\mathbb{C}^n$ with open unit ball B, and put*

$$V = \sup_{z\in B} \Big(\sum_{|S|=m} |z^S|^2\Big)^{1/2}.$$

Then there are signs $\varepsilon_S = \pm 1$ such that

$$\sup_{z\in B} |P_\varepsilon(z)| \le 4V\sqrt{2n\log(1+4em) + \log 8}.$$

Proof. Let (ε_S) be independent Rademacher random variables, and put $R = 2V\sqrt{2n\log(1+4em) + \log 8}$. Fix z in the closure of B. For real numbers a_S, we have $\mathbb{P}(|\sum_S \varepsilon_S a_S| > t) \le 2\exp(-t^2/(2\sum_S a_S^2))$ for $t > 0$ [9, (17.3)]. If $|P_\varepsilon(z)| > R$, then $|\operatorname{Re} P_\varepsilon(z)|$ or $|\operatorname{Im} P_\varepsilon(z)|$ exceeds $R/\sqrt{2}$. We apply the tail bound with $a_S = \operatorname{Re} z^S$ and with $a_S = \operatorname{Im} z^S$ (note that $\operatorname{Re} P_\varepsilon(z) = \sum_S \varepsilon_S \operatorname{Re} z^S$, since the ε_S are real). In both cases $\sum_S a_S^2 \le \sum_S |z^S|^2 \le V^2$, by continuity also on the closure of B, and we find that

$$\mathbb{P}\big(|P_\varepsilon(z)| > R\big) \le 4e^{-R^2/(4V^2)}.$$

By [9, Lemma 17.2] with radius $1/(2em)$, there is a set F in the closure of B with $|F| \le (1+4em)^{2n}$ such that every point of B lies within distance $1/(2em)$ of F. The union bound gives

$$\mathbb{P}\Big(\max_{v\in F}|P_\varepsilon(v)| > R\Big) \le 4(1+4em)^{2n} e^{-R^2/(4V^2)} = \frac{1}{2}.$$

Hence there are signs with $|P_\varepsilon| \le R$ on F. Every m-homogeneous polynomial Q satisfies

$$|Q(z)-Q(v)| \le em\|z-v\| \sup_B |Q|$$

for z, v in the closure of B. Indeed, by [9, (17.9)] the left-hand side is at most $m\|z-v\| \sup_{u,w}|\check{Q}(u,w,\dots,w)|$, and Harris' polarization estimate [9, (2.51)] with $n_1 = 1$ and $n_2 = m-1$ bounds the supremum by $(m/(m-1))^{m-1}\sup_B|Q| \le e\sup_B|Q|$; see [9, p. 441]. For $z \in B$ we choose $v \in F$ with $\|z-v\| \le 1/(2em)$ and obtain

$$|P_\varepsilon(z)| \le |P_\varepsilon(v)| + |P_\varepsilon(z)-P_\varepsilon(v)| \le R + \frac{1}{2}\sup_B|P_\varepsilon|.$$

Taking the supremum over $z \in B$ gives $\sup_B|P_\varepsilon| \le 2R$. □

Combining Lemmas 1 and 2 for $B = B_p^n$, we find that for every $1 \le p \le 2$ and $1 \le m \le n$ there are signs $\varepsilon_S = \pm 1$ such that

$$\sup_{z\in B_p^n}|P_\varepsilon(z)| \le 4\sqrt{2n\log(1+4em)+\log 8}\;\frac{m^{-m(1/p-1/2)}}{\sqrt{m!}}. \tag{4}$$

3. Proof of Theorem 1

Fix $1 < p < 2$ and put $\sigma = 1 - 1/p$, so that $s = 2\sigma$. Let $2 \le m < n$, and let P_ε be as in (4). We first bound $K(B_p^n)$ for fixed n and m, and then choose m.

Suppose that $0 < r < K(B_p^n)$. If the Bohr inequality holds on $r'B_p^n$, then it holds on $rB_p^n \subset r'B_p^n$ for $r < r'$. Hence it holds on rB_p^n: for every $f \in H_\infty(B_p^n)$ and every $z \in rB_p^n$ we have $\sum_\alpha |c_\alpha(f)z^\alpha| \le \|f\|_\infty$. We apply this to $f = P_\varepsilon$ and to the point $z = \rho r n^{-1/p}(1,\dots,1)$, where $0 < \rho < 1$, so that $\|z\|_p = \rho r < r$. Since P_ε has $\binom{n}{m}$ coefficients of modulus 1, we obtain $\binom{n}{m}(\rho r)^m n^{-m/p} \le \|P_\varepsilon\|_\infty$. Letting $\rho \to 1$ gives

$$r^m \le \binom{n}{m}^{-1} n^{m/p}\|P_\varepsilon\|_\infty.$$

We now insert (4), the bound $\binom{n}{m} \ge (n-m)^m/m!$, and the inequality $\sqrt{m!} \le e^{1/2}m^{1/4}(m/e)^{m/2}$, which follows from $m! \le e\sqrt{m}(m/e)^m$. With

$A_{n,m} = 4e^{1/2} m^{1/4} \sqrt{2n\log(1+4em) + \log 8}$, we find that

$$\begin{aligned} r^m &\le 4\sqrt{2n\log(1+4em)+\log 8}\, m^{-m(1/p-1/2)}\sqrt{m!}\,\frac{n^{m/p}}{(n-m)^m} \\ &\le A_{n,m}\, e^{-m/2}\, m^{-m(1/p-1/2)+m/2}\, n^{-m\sigma}\Big(\frac{n}{n-m}\Big)^m \\ &= A_{n,m}\, e^{-m/2}\Big(\frac{n}{n-m}\Big)^m\Big(\frac{m}{n}\Big)^{m\sigma}, \end{aligned}$$

where we used $n^{m/p}(n-m)^{-m} = n^{-m\sigma}(n/(n-m))^m$ and $-(1/p-1/2)+1/2 = \sigma$. Letting $r \to K(B_p^n)$ and taking m-th roots, we obtain, for every $2 \le m < n$,

$$K(B_p^n) \le A_{n,m}^{1/m}\, e^{-1/2}\,\frac{n}{n-m}\Big(\frac{m}{n}\Big)^{\sigma}. \tag{5}$$

This is (3) with $\gamma = 1/2$, since $A_{n,m} = n^{1/2}$ up to factors that are polynomial in $\log n$ when $m \asymp \log n$.

We now choose m. Fix $t > 0$ and put $m = \lceil t\log n\rceil$; then $2 \le m < n$ for all large n. Since

$$\begin{aligned} \log A_{n,m} &= \log(4e^{1/2}) + \frac{1}{4}\log m + \frac{1}{2}\log\big(2n\log(1+4em)+\log 8\big) \\ &= \frac{1}{2}\log n + O(\log\log n) \end{aligned}$$

and $m = t\log n + O(1)$, we have $(\log A_{n,m})/m \to 1/(2t)$, that is, $A_{n,m}^{1/m} \to e^{1/(2t)}$. Moreover $n/(n-m) \to 1$ and $m/\log n \to t$. Dividing (5) by $((\log n)/n)^\sigma$, we find that

$$\limsup_{n\to\infty} \frac{K(B_p^n)}{((\log n)/n)^\sigma} \le t^\sigma \exp\left(\frac{1}{2t} - \frac{1}{2}\right).$$

The logarithm of the right-hand side is $\varphi(t) = \sigma\log t + 1/(2t) - 1/2$, and $\varphi'(t) = \sigma/t - 1/(2t^2)$ vanishes only at $t = 1/(2\sigma) = 1/s$, where φ attains its minimum

$$\varphi(1/s) = -\frac{s}{2}\log s + \frac{s}{2} - \frac{1}{2} = -\frac{1-s+s\log s}{2}.$$

This is the claimed bound. □

In other words, the proof uses degree $m \approx (\log n)/s = p\log n/(2(p-1))$, which is larger than the degree $m \approx \log n$ used in the proof of (1), and tends to infinity relative to $\log n$ as $p \to 1$.

4. Concluding remarks

Numerical values. The constant κ_p is close to 1 unless p is close to 1. For instance, $\kappa_{3/2} = 0.9689\ldots$, $\kappa_{5/4} = 0.8898\ldots$ and $\kappa_{11/10} = 0.7756\ldots$. As $p \to 1$ we have $s \to 0$ and $\kappa_p \to e^{-1/2}$. The bound (5) is explicit, but it approaches κ_p slowly; numerically, at $p = 3/2$ and $n = 10^{12}$ the right-hand side of (5), minimized over m, is about 1.06.

The case $p \geq 2$. For $p = 2$ we have $s = 1$ and $\kappa_2 = 1$, so Theorem 1 is consistent with (2). For $p > 2$ the exponent $2/p - 1$ in the proof of Lemma 1 is negative, the variance bound fails, and our argument gives nothing new; there (2) holds anyway. Indeed, for $p \geq 2$ the flat point maximizes the variance as well as the majorant, so the mismatch exploited in the proof is absent.

Kahane–Salem–Zygmund inequalities with the pointwise variance. Defant and Mastyło [10] obtained Kahane–Salem–Zygmund inequalities for subgaussian random polynomials on arbitrary finite dimensional Banach spaces. For Rademacher variables and exponent $r = 2$ (Theorem 6.9 in the arXiv version of [10]), their bound for P_ε is a constant times $(n(1 + \log m))^{1/2}V$, with V as in Lemma 2. Since only the order of this factor enters the proof of Theorem 1, their inequality could replace Lemma 2; the new ingredient is Lemma 1.

The exact asymptotics. Theorem 1 leaves open the exact behaviour of $K(B_p^n)$. We have not tried to optimize κ_p. For instance, Lemma 1 is not sharp: numerically, the supremum of $\sum_S |z^S|^2$ over B_p^n is attained at points with roughly $2m$ equal non-zero coordinates (for $p = 3/2$), where it is smaller than the bound in Lemma 1 by a factor of about 0.59^m.

Problem 1. *Let $1 < p < 2$. Does $K(B_p^n)/((\log n)/n)^{1-1/p}$ converge as $n \to \infty$? Determine its limit inferior and limit superior.*

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University of Oslo, Oslo, Norway
Email address: oistein.sovik@usit.uio.no